

Ad Creative Similarity & Relationship Discovery Report

1. Overview & Architecture:

1.1 Problem Statement

Discovering and classifying pairwise creative relationships between display advertisement images from the PeterBrendan/AdImageNet dataset.

1.2 Relationship Taxonomy

The system classifies candidate creative ad pairs into 6 relationship categories:

1. **Identical:** same creative (allowing for recompression/near-duplicate noise).
2. **Containment:** one image is a crop/resize of the other (e.g. same creative at a different ad size)
3. **Color-variant:** same layout and content, only the color/palette differs
4. **Text-variant:** same layout and visual content, only the overlaid text differs
5. **Layout-variant:** same template/structure, different photo or content.
6. **Unrelated:** no meaningful relationship

1.3 Multi-Modal Signals Extracted

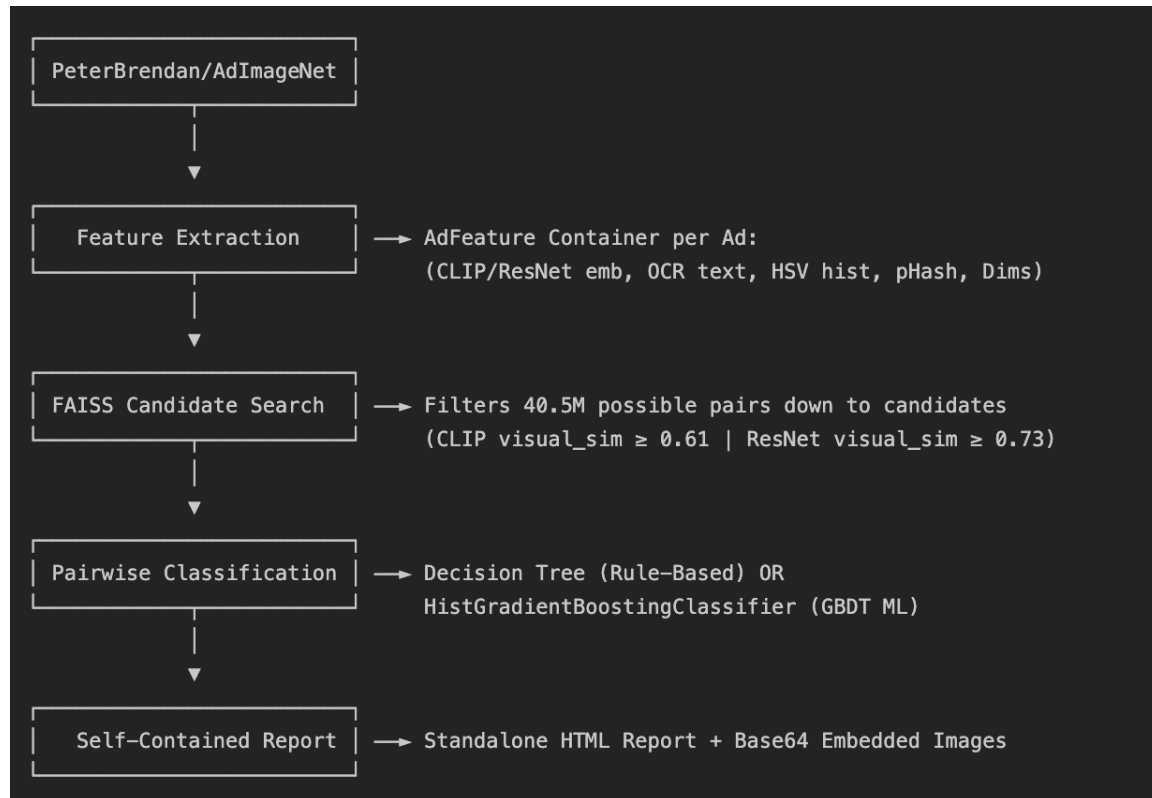
Each candidate pair (ad1, ad2) is represented by 5 feature signals:

1. **visual_sim:** Cosine similarity of global visual embeddings (CLIP ViT-B/32 vs. ResNet-18 baseline).
2. **text_sim:** OCR text similarity using Sentence Transformers (all-MiniLM-L6-v2) vs. Lexical Jaccard.
3. **color_sim:** HSV canvas color histogram correlation.
4. **phash_dist:** Perceptual hash Hamming distance for near-duplicate detection.
5. **same_dim:** Boolean flag indicating whether two ads share matching standard IAB aspect ratio dimensions or identical raw pixel dimensions

1.4 Approach and Execution Workflow:

The *end-to-end execution workflow* connects dataset ingestion, multi-modal feature extraction, scalable candidate retrieval, and pairwise relationship classification into an automated pipeline. By leveraging FAISS vector indexing over *global visual embeddings*, the system reduces over 40 million potential ad combinations down to a targeted set of high-confidence candidate pairs within seconds. These candidate pairs are then evaluated using either calibrated rule-based decision trees or a trained Gradient Boosted Decision Tree (GBDT) classifier to assign exact relationship taxonomy labels.

Finally, all pairwise matches, summary statistics, and side-by-side visual comparisons are compiled into a self-contained HTML report with Base64-embedded image assets for an offline review.



2. Experiment Results and Analysis:

All five pipeline configurations were evaluated on the full **9,003** ad creative dataset 40.5M candidate pairs evaluated via FAISS:

2.1 Comparative Results Matrix

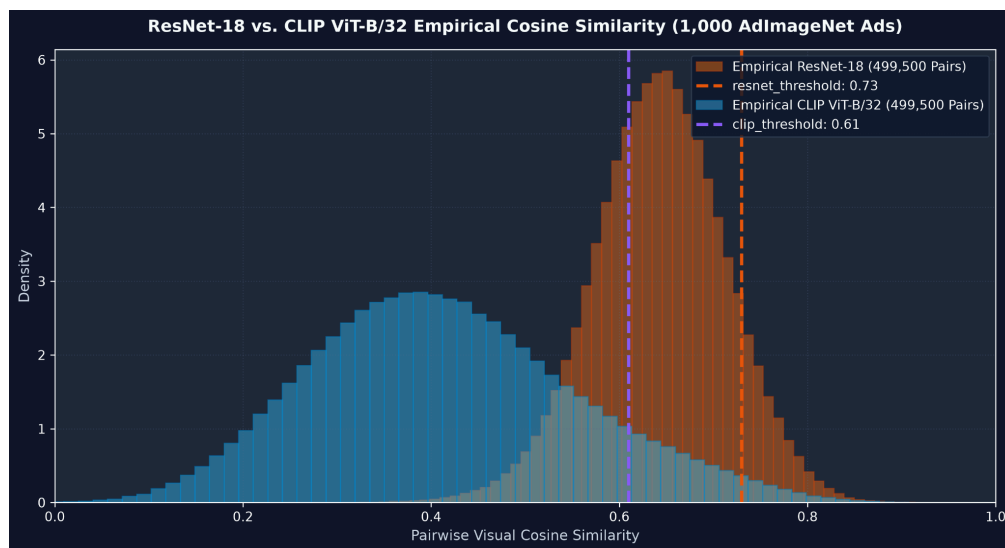
No.	Pipeline Configuration	Related ads	Identical	Containment	Color-variant	Text-variant	Layout-variant	Time (on MPS)
1	Rule-Based (ResNet-18 + Lexical Jaccard)	2335 / 9003 (25.9%)	556	1719	20	627	604	2m 24s
2	Rule-Based (CLIP + Sentence Transformers)	1975 / 9003 (21.9%)	548	1514	66	358	1089	4m 58s
3	Rule-Based (CLIP + Lexical Jaccard)	1870 / 9003 (20.8%)	549	1119	70	289	812	3m 39s

4	GBDT ML Classifier (CLIP + Sentence Transformers)	3664 / 9003 (40.7%)	551	2987	779	437	3261	5m 46s
5	GBDT ML Classifier (CLIP + Lexical Jaccard)	2830 / 9003 (31.4%)	633	2623	642	482	976	4m 32s

Deciding factors for the best classification solution:

1. Resnet18 vs CLIP:

- CLIP → meaningful visual_sim scores for ad creatives.
- Resnet18 → High False Positive Risk (as seen below in the figure).



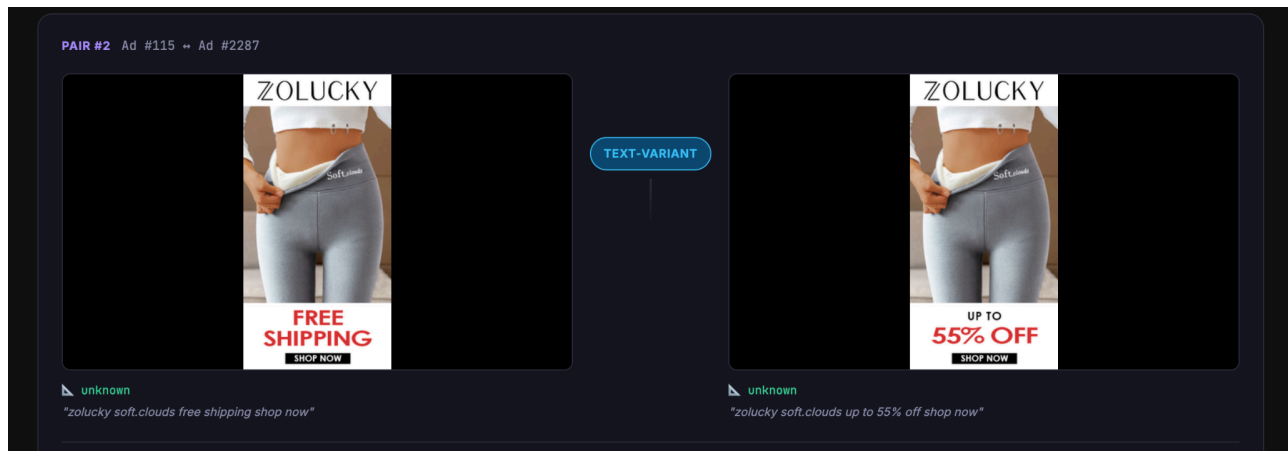
2. Jaccard vs Sentence Transformer:

- Jaccard → strong baseline accuracy for exact-copy ads.
- Sentence Transformer → captures synonyms, paraphrasing, context. Hence, better for generalisation.

Example: Ad 115 - Ad 2287

Jaccard similarity is 0.455 drops due to OCR noise

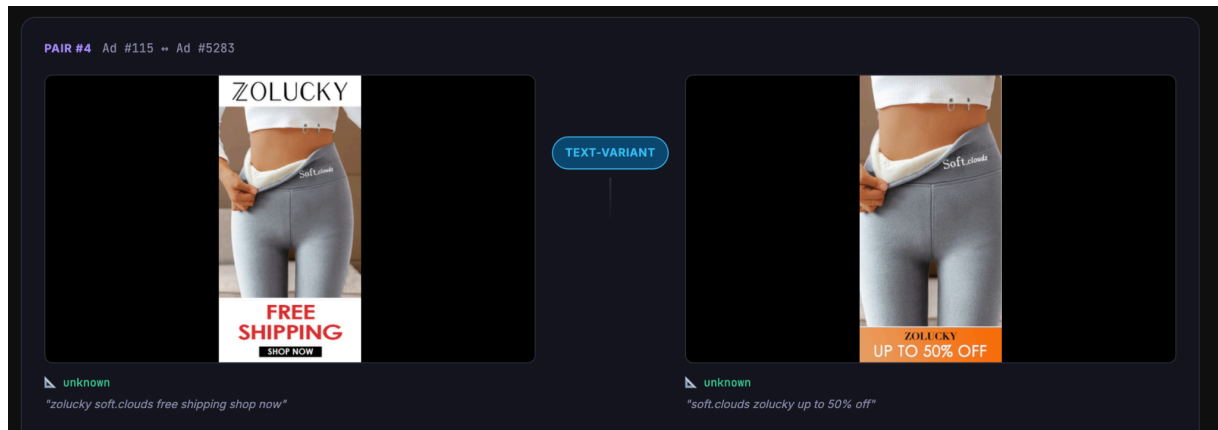
Sentence Transformer cosine similarity is 0.803.



3. Rule based vs GBDT (gradient boost decision tree):

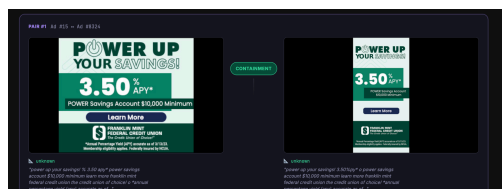
- Rule based → strict boundary; high precision on clean pairs but can suffer from lower recall
- GBDT → more dataset relationship discovery but prone to FP due to Limitation of Global Visual Embeddings.

Ex: GBDT needs better visual_sim embeddings specific to layout to solve for FP in text-variant.

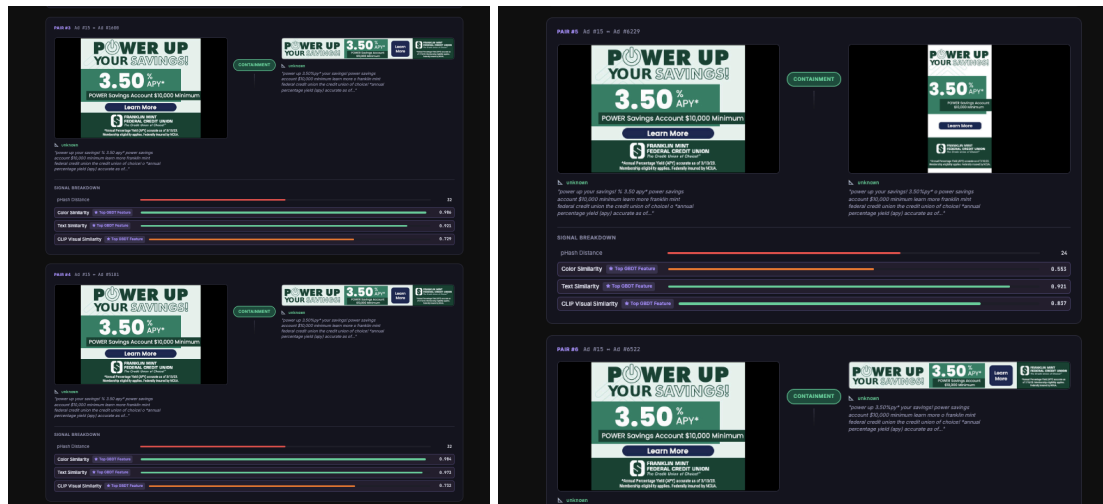


There are 2 contenders based on above observations/analysis:

- Rule-Based (CLIP + Sentence Transformers)** → better precision and captures clean pairs.



2. **GBDT ML Classifier (CLIP + Lexical Jaccard)** → Discovers the **highest number of creative relationships** across the dataset (for now choosing Lexical Jaccard due to more FP in sentence transformer coming from GBDT training dataset quality issue)



4. Future Directions (Next Steps)

With more time, I would focus on the following:

1. **Layout-Specific Visual Embeddings:** Replace global CLIP pooling vectors with layout-aware visual embeddings (e.g., LayoutLM spatial bounding box tokens, DINOv2 patch features, or SAM object masking) to eliminate false positive Text-variants in the GBDT classifier.
2. **Fine-Tuned Multi-Modal Metric Learning:** Train a multi-modal PyTorch projection head using Triplet / Contrastive Loss directly on labeled ad pairs to learn a unified joint embedding space for creative relationships.
3. **Classifier & Dataset Quality Enhancements:**
 - a. **Dataset Quality & Active Learning:** Clean pseudo-labeling noise using Active Learning (sampling uncertain pairs near 0.50 probability) and synthetic data augmentations (simulated OCR typos, color shifts, aspect crops).
 - b. **Classifier Alternative (Graph Neural Networks):** Replace isolated pairwise GBDT predictions with Relational GNNs (R-GCN) to model campaign clusters as connected graphs and enforce relational transitivity.